- The number of Barr bodies follows N-1 rule (N minus one rule), where N is the total number of X chromosomes present.

Haplodiploidy in Honeybees

In hymenopteran insects such as honeybees, ants and wasps a mechanism of sex determination called haplodiploidy mechanism of sex determination is common. In this system, the sex of the offspring is determined by the number of sets of chromosomes it receives. Fertilized eggs develop into females (Queen or Worker) and unfertilized eggs develop into males (drones) by parthenogenesis. It means that the males have half the number of chromosomes (haploid) and the females have double the number (diploid), hence the name haplodiplody for this system of sex determination.

This mode of sex determination facilitates the evolution of sociality in which only one diploid female becomes a queen and lays the eggs for the colony. All other females which are diploid having developed from fertilized eggs help to raise the queen's eggs and so contribute to the queen's reproductive success and indirectly to their own, a phenomenon known as **Kin Selection**. The queen constructs their social environment by releasing a hormone that suppresses fertility of the workers.

4.5 Sex Linked Inheritance



The inheritance of a trait that is determined by a gene located on one of the sex chromosomes is called sex linked inheritance. Genes present on the differential region of X or Y chromosomes are called sex linked genes. The genes present in the differential region of X chromosome are called X linked genes. The X–linked genes have no corresponding alleles in the Y chromosome. The genes present in the differential region of Y chromosome are called Y-linked or **holandric genes**. The Y linked genes have no corresponding allele in X chromosome. The Y linked genes inherit along with Y chromosome and they phenotypically express only in the male sex. Sex linked inherited traits are more common in males than females because, males are hemizygous and therefore express the trait when they inherit one mutant allele. The X – linked and Y – linked genes in the differential region (non–homologus region) do not undergo pairing or crossing over during meiosis. The inheritance of X or Y linked genes is called sex-linked inheritance.

4.5.1 Inheritance of X - linked genes

Red-green colour blindness or daltonism, haemophilia and Duchenne's muscular dystrophy are examples of X-linked gene inheritance in humans.

1. Haemophilia

Haemophilia is commonly known as bleeder's disease, which is more common in men than women. This hereditary disease was first reported by John Cotto in 1803. Haemophilia is caused by a recessive X-linked gene. A person with a recessive gene for haemophilia lacks a normal clotting substance (thromboplastin) in blood, hence minor injuries cause continuous bleeding, leading to death. The females are carriers of the disease and would transmit the disease to 50% of their sons even if the male parent is normal. Haemophilia follows the characteristic criss - cross pattern of inheritance.

2. Colour blindness

In human beings a dominant X – linked gene is necessary for the formation of colour sensitive cells, the cones. The recessive form of this gene is incapable of producing colour sensitive cone cells. Homozygous recessive females (X^cX^c) and hemizygous recessive males (X^cY) are unable to distinguish red and green colour. The inheritance of colour blindness can be studied in the following two types of marriages.

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